MODULE 07 · SECTION 1 - DESIGNING DATA PROCESSING SYSTEMS

Migrating Files and Bulk Data

gcloud storage, Storage Transfer Service, Transfer Appliance, and the network underneath them.

SECTION 1 of 5 EXAM WEIGHT 22% MODULE 7 of 45

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Every bulk migration starts with one sum: **size divided by bandwidth.**

What Larkspur has to move

DATA	SIZE	THE CATCH
Reading history in HDFS	1.2 PB of ORC, 8 years	2.9 million small files
New readings, until cutover	About 30 GB a day	The history keeps growing
Engineers' manuals	18,000 PDF pages	Small, one-off copy
Westgrid settlement files	Daily, in Amazon S3	Data in another cloud

THE CONSTRAINT

The Bristol lease ends on 30 June 2027, and today's link is 1 Gbps.

A terabyte a day for every 100 Mbps

LINK	PER DAY, FLAT OUT	1.2 PB TAKES
100 Mbps	About 1 TB	About 3 years
1 Gbps, Larkspur today	About 10 TB	Almost 4 months
10 Gbps, Interconnect on order	About 100 TB	About 11 days
Five Transfer Appliances	No link needed	Weeks, plus shipping

HONEST LIMIT

These are best cases. Real links rarely run flat out, and the business still needs its link.

Google's starting point

WHERE THE DATA IS	UNDER 1 TB	OVER 1 TB
On-premises	gcloud storage	Storage Transfer Service or Transfer Appliance
Another Cloud Storage region	gcloud storage	Storage Transfer Service
Another cloud, such as S3 or Azure	Storage Transfer Service	Storage Transfer Service

THE TIEBREAKER

Over the network would take more than about a week? Google suggests Transfer Appliance.

gcloud storage: the command line

- **For small, one-off copies** and syncs: **cp**, **mv** and **rsync**.
- **Checksums are automatic.** MD5 or CRC32C are compared, and a bad copy is deleted.
- **Parallel composite uploads** split files over 150 MiB into up to 32 chunks.
- **Composite objects have no MD5**, and the trick is skipped with retention policies.

> CURRENCY CHECK
>
> gsutil is legacy, and leaves the Google Cloud CLI after March 2027. Answers should say gcloud storage.

gcloud storage in practice

```
# One-off copy of the manuals, then rerun to pick up edits
gcloud storage rsync --recursive /srv/manuals gs://lk-manuals

# Big exports: parallel composite uploads, Standard class only
gcloud config set storage/parallel_composite_upload_enabled True
gcloud storage cp ./exports/*.orc gs://lk-history-land/

# Spot-check: compare the local hash with the object's hash
gcloud storage hash ./exports/part-0001.orc
gcloud storage objects describe gs://lk-history-land/part-0001.orc
```

2 A rerun copies only new or changed files.

5-6 Files over 150 MiB go up in parallel chunks.

9-10 A composite object shows a CRC32C, but no MD5.

HOW IT WORKS

Every copy is already checked automatically. The spot-check is evidence for your records.

Storage Transfer Service

- **Managed jobs**, once or on a schedule. Retries, logging and monitoring built in.
- **Agentless** from Cloud Storage, Amazon S3, Azure Blob and URL lists.
- **Agents** for file systems, **HDFS** and S3-compatible storage. At least three per pool.
- **Incremental**: compares time and size, copies what changed. **Bandwidth** can be capped.

> HOW IT WORKS
>
> Agents run next to the data. Add more agents, and the transfer speeds up.

Agents in Bristol, bucket in London

01

Install agents

Containers on machines that can read the Hadoop cluster.

02

Form a pool

Three or more agents, same access, bandwidth capped.

03

Compare

The service checks both sides and finds what changed.

04

Copy and check

Agents read, checksum and send. A mismatch fails.

05

Land

Objects arrive in a europe-west2 bucket, with logs.

FOR LARKSPUR

Run it nightly for new days, then once more at cutover with writes frozen.

Scheduled or event-driven?

SCHEDULED

List, compare, copy

- Runs once, or on a timetable
- Lists the source on every run
- Can delete extra destination files
- Right for batches and migrations

EVENT-DRIVEN

Listen, then copy

- Listens for change notifications
- S3 via SQS, Azure via queues, Pub/Sub
- No listing, so near real time
- Deletions are not carried across

THE TIEBREAKER

'Within minutes of arrival' points to event-driven. 'Mirror deletions' points to scheduled.

Newer transfer options

- **Google-managed private network** for S3 sources: **no AWS internet egress charges**.
- Instead, a **per-GiB fee** to Google. Capacity is shared, and can slow at peak times.
- **Transfer Appliance online mode** streams to your bucket without shipping the box.
- **Cross-Cloud Interconnect**: a dedicated link to AWS, Azure, Oracle or Alibaba.

> THE SIGNAL
>
> 'Cut the egress bill from AWS' is a signal for the managed private network.

Transfer Appliance: a van full of disks

- **Google ships you a storage server.** Copy over NFS, seal it, and ship it back.
- **AES-256 encrypted**, with your key in Cloud KMS. Wiped to NIST 800-88 afterwards.
- **Up to 300 TB each.** Order several, and fill them in parallel.
- Google's example: **300 TB in under 25 days**, then about **10 business days** to land.

> THE SIGNAL
>
> Use it when the network would take more than about a week, or there is no good network.

2.9 million small files

- **Small files are slow everywhere.** Per-file overhead dominates, whatever the link.
- Transfer Appliance **rehydration slows down** with very small files.
- Storage Transfer Service: keep source folders **under about 1 million files**.
- **Compact before you move.** Fewer, larger ORC files copy faster and query faster.

> THE TAKEAWAY
>
> Fixing small files at the source helps the transfer now, and Spark and BigQuery later.

Four ways into Google Cloud

OPTION	CAPACITY	ENCRYPTED?	PICK IT WHEN
Public internet	Whatever you buy	Only by TLS in apps	Small, quick, cheap
HA VPN	1 to 3 Gbps per tunnel	Yes, IPsec	Modest bandwidth, encrypted
Dedicated Interconnect	10, 100 or 400 Gbps circuits	Not by default	You reach a colocation site
Partner Interconnect	50 Mbps to 50 Gbps	Not by default	A provider reaches Google

REMEMBER

HA VPN gives a 99.99% SLA with the right topology. A single interface gives none.

Private is not encrypted

HA VPN OVER INTERCONNECT

IPsec, end to end

- Layer 3 encryption on the private link
- Traffic never touches the internet
- Covers the provider's network too
- Several tunnels to fill a big link

MACSEC

Encrypts one hop

- Layer 2, IEEE 802.1AE
- Your router to Google's edge
- Partner: the provider's leg is separate
- Google advises adding IPsec too

THE TRAP

Interconnect keeps traffic off the internet. It does not encrypt it.

Validate before you delete the source

- **Checksums in flight.** Storage Transfer Service checks end to end, and fails mismatches.
- **Count and compare** afterwards: files, bytes and checksums, folder by folder.
- **Freeze writes** for the final pass, or a changing file may land half-written.
- **Delete the source last.** The service cannot restore what you removed.

> FOR LARKSPUR
>
> The lease ends on 30 June 2027. Plan the final sync and its checks well before then.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
A few hundred GB, one-off, from a server	gcloud storage
Petabytes, slow or shared link	Transfer Appliance
From Amazon S3 or Azure, recurring	Storage Transfer Service
On-premises **NFS or HDFS**, good network	Storage Transfer Service with agents
Within minutes of new objects arriving	Event-driven transfers
Encrypted over Interconnect	HA VPN over Interconnect, or MACsec

THE PATTERN

Size, source and speed decide most of these questions.

> Larkspur must move 1.2 PB of frozen Hive history from Bristol before the lease ends on 30 June 2027. The 1 Gbps internet link also carries the company's daily traffic.
>
> A 10 Gbps Partner Interconnect is on order, but the provider has not given an installation date. New days keep arriving until cutover.

What should the team do?

A Copy the history with gcloud storage rsync from an edge node, running at night

B Use Storage Transfer Service agents over the 1 Gbps link, with a bandwidth cap

C Wait for the Partner Interconnect, then run Storage Transfer Service across it

D Ship Transfer Appliances for the history, then sync new days with agents

◆ PAUSE AND DECIDE

What should the team do?

A gcloud storage rsync at night

A small-job tool for a petabyte. Running only at night, it would take about a year.

B Agents over the 1 Gbps link

The right tool on the wrong link. Capped to protect the business, it runs for many months.

C Wait for the Interconnect

It bets a fixed deadline on an installation date that nobody has given.

D Appliances, then agents

The bulk travels by road in weeks. Agents copy each new day, with a final pass at cutover.

THE SIGNAL

Do the division first. Months over the network, against a fixed deadline, means the van.

> The field engineer assistant needs Larkspur's 18,000 pages of manuals in Cloud Storage. They total under 50 GB, on one file server in Bristol.
>
> It is a one-off copy, with occasional edits afterwards. The team wants the simplest option that checks what it copies.

Which approach fits best?

A. Run gcloud storage rsync from the file server, and rerun it after edits

B. Deploy a pool of three Storage Transfer Service agents next to the server

C. Order a Transfer Appliance, and copy the manuals onto it over NFS

D. Use the BigQuery Data Transfer Service to load the files into a dataset

◆ PAUSE AND DECIDE

Which approach fits best?

A gcloud storage rsync

Under a terabyte, one-off, with automatic checksums. A rerun copies only the edits.

B Three transfer agents

It works, but it means containers, a pool and jobs, for a copy that takes minutes.

C A Transfer Appliance

Days of shipping, for data that crosses a 1 Gbps link in about five minutes.

D BigQuery Data Transfer Service

It loads BigQuery tables, and cannot read a file server. The assistant needs the files.

THE SIGNAL

Under a terabyte, from one machine, is gcloud storage. 'Simplest' is a requirement, not a hint.

> Westgrid writes settlement files into an Amazon S3 bucket throughout the day. Larkspur needs each file in Cloud Storage within minutes of it arriving.
>
> The bucket already holds millions of older files, and Westgrid never deletes anything.

How should Larkspur copy the files?

A. A VM running gcloud storage rsync from the S3 bucket every five minutes

B. A Storage Transfer Service job from S3, scheduled to run every hour

C. An event-driven Storage Transfer Service job, fed by S3 notifications

D. BigQuery Data Transfer Service for Amazon S3, loading a settlement table

◆ PAUSE AND DECIDE

How should Larkspur copy the files?

A rsync from a VM, every five minutes

Each run lists millions of files, and the VM is yours to patch and watch.

B A scheduled hourly job

Managed, but an hour is not minutes, and every run lists the whole bucket.

C An event-driven transfer

SQS notifications trigger each copy. No listing, near real time, no deletes to miss.

D BigQuery DTS for Amazon S3

It loads tables on a schedule. Larkspur asked for the files in Cloud Storage.

THE SIGNAL

'Within minutes' plus millions of files to list points to events, not schedules.

> The 10 Gbps Partner Interconnect is finally live, and Larkspur will send new data and the final sync across it.
>
> Security requires all traffic between Bristol and Google Cloud to be encrypted at the network layer, including across the provider's network.

What should the network team do?

A Nothing extra, because Interconnect traffic never crosses the internet

B Run HA VPN tunnels over the Interconnect, enough of them to fill the link

C Replace the Interconnect with HA VPN tunnels over the 1 Gbps internet link

D Enable MACsec on the Interconnect, and rely on it as the only encryption

◆ PAUSE AND DECIDE

What should the network team do?

A. Nothing extra

Private is not encrypted. Off the internet is not the same as protected.

B. HA VPN over Interconnect

IPsec runs end to end, across the provider too. Several tunnels fill the link.

C. HA VPN over the internet

Encrypted, but capped by a 1 Gbps link while the Interconnect sits idle.

D. MACsec alone

It covers the hop to Google's edge. The provider's network is left out.

THE SIGNAL

'Including the provider's network' rules out MACsec alone. IPsec over Interconnect covers every hop.

Module 07 in five lines

- Do the division first: **100 Mbps moves about 1 TB a day**.
- **gcloud storage** for small copies. **Storage Transfer Service** for big, recurring or cross-cloud.
- **Agents** for file systems and HDFS. **Event-driven** for minutes. **Scheduled** to mirror deletes.
- **Transfer Appliance** when the network would take over a week. Compact small files first.
- **Private is not encrypted.** Add IPsec or MACsec, then validate before deleting.

COMING UP

Next: module 08, migrating databases and warehouses.